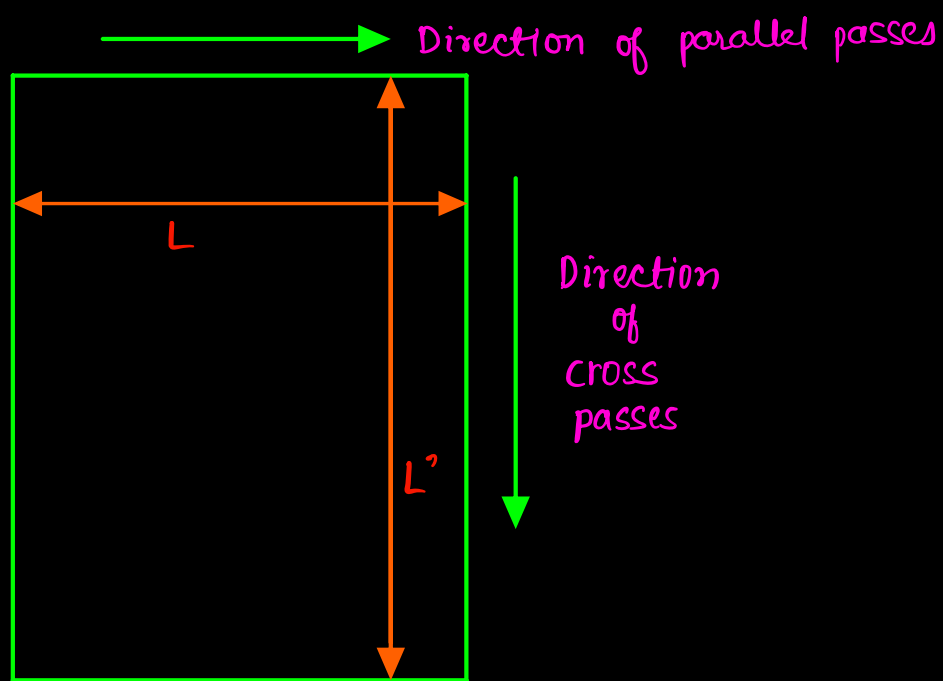




V_r = speed of ripper in top gear

L = length of l^{th} pass

L' = length of x -pass

T = Time of ripper to travel back to the next furrow (in seconds) during l^{th} passes

T' = Time of ripper to travel back to the next furrow (in seconds) during x -passes

C = Spacing between two passes in l^{th} pass

C' = spacing between two passes in x -pass

Now,

$$\text{No. of passes req. in } l^{th} \text{ pass} = \left(\frac{L'}{C} \right)$$

$$\text{No. of passes req. in } x\text{-pass} = \left(\frac{L}{C'} \right)$$

$$\text{Total time in } l^{th} \text{ passes} = \left[\frac{L'}{C} \times \frac{L}{V_r} + T \times \frac{L'}{C} \right]$$

$$\text{Total time in } x\text{-passes} = \left[\frac{L}{C'} \times \frac{L'}{V_r} + T' \times \frac{L}{C'} \right]$$

$$\begin{aligned} \text{Total time for both } 11^{\text{el}} \text{ \& } x\text{-passes} &= \left[\frac{L'}{C} \times \frac{L}{V_r} + \frac{T L'}{C} \right] + \left[\frac{L}{C'} \times \frac{L'}{V_r} + \frac{T' L}{C'} \right] \\ &= \frac{L L'}{V_r} \left[\frac{1}{C} + \frac{1}{C'} \right] + \left(\frac{T L'}{C} + \frac{T' L}{C'} \right) \end{aligned}$$

Now,

Total vol. excavated in 11^{el} \& x -passes = $h_t \times L \times L'$ m^3
 If L' is chosen such that ($T = T'$);

Then,

$$\text{Theoretical productivity } \left(\frac{\text{m}^3}{\text{s}} \right) = \frac{h_t \times L \times L'}{\frac{L L'}{V_r} \left(\frac{1}{C} + \frac{1}{C'} \right) + \left(\frac{T L'}{C} + \frac{T' L}{C'} \right)}$$

$$L = L' \Rightarrow T = T'$$

$$Q_{\text{th}} = \frac{h_t \times \cancel{L} \times \cancel{L'}}{\frac{\cancel{L} \cancel{L}'}{V_r} \left(\frac{1}{C} + \frac{1}{C'} \right) + \cancel{L'} T \left(\frac{1}{L C} + \frac{1}{L' C'} \right)}$$

$$= \frac{h_t}{\frac{1}{V_r} \left(\frac{1}{C} + \frac{1}{C'} \right) + T \left(\frac{1}{L C} + \frac{1}{L' C'} \right)}$$

Practical

Hourly productivity considering, K_u (ripper utilization factor)

$$K_u = 0.7 \text{ to } 0.8$$

$$Q_r(11^{\text{el}} + x) = \frac{3600 \times h_t \times K_u}{\frac{1}{V_r} \left(\frac{1}{C} + \frac{1}{C'} \right) + T \left(\frac{1}{L C} + \frac{1}{L' C'} \right)} \text{ m}^3/\text{hr}$$

★ Productivity calculation of 11^{el} passes only —

$$\text{No. of passes in } 11^{\text{el}} = \frac{L'}{C}$$

$$\text{Volume loosened in } 11^{\text{el}} \text{ passes} = (h_{\text{eff}} \times L \times L')$$

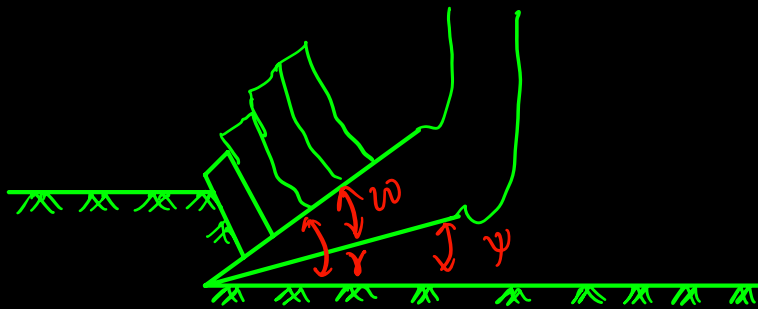
$$\text{Total time taken in parallel passes} = \left(\frac{L'}{C} \times \frac{L}{v_r} + T \times \frac{L'}{C} \right)$$

$$Q \left(\frac{m^3}{s} \right) = \frac{h_{eff} \times L \times L'}{\left(\frac{LL'}{v_r C} + \frac{T L'}{C} \right)}$$

$$= \frac{h_{eff}}{\frac{1}{C} \left(\frac{1}{v_r} + \frac{T}{L} \right)}$$

Hourly productivity considering effective ripper utilisation factor
 $K_u = 0.7 \sim 0.8$

$$Q_r(11el) = \frac{h_{ef} \times 3600 \times K_u}{\frac{1}{C} \left(\frac{1}{v_r} + \frac{T}{L} \right)}$$



ω = tip angle ($20^\circ + 30^\circ$)

γ = cutting angle (

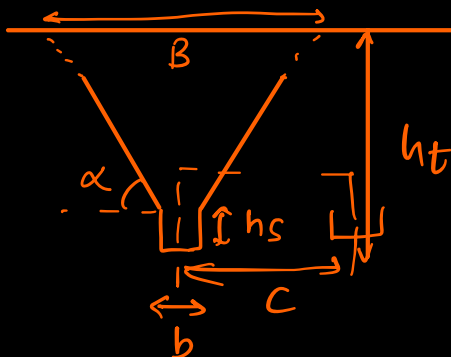
ψ = tooth clearance angle ($5^\circ \text{ to } 7^\circ$ / $8^\circ \text{ to } 10^\circ$)

igneous/metamorphic

frozen rock mass

γ is automatically derived from $(\psi + \omega)$

⇒ Smaller clearance angle cause crushing of rock mass



slot width = b

slot height = h_s (0.15 to 0.2 of h_t)

slope of sidewall = α ($40^\circ - 60^\circ$)

* Rock mass loosening ability of ripper is specified by

→ possible penetration (h_t) of ripper tooth into loosened rock-mass

Monolithic rock mass → loosening due to overcoming tensile strength

Jointed rock mass → overcoming cohesion b/w str. block

C = distance between adjacent passes

→ set with an aim to ensure req. lump size

→ sufficient loosening depth

$$\frac{h_t - h_s}{x} = \tan \alpha$$

$$\frac{y}{\frac{(C-b)}{2}} = \tan \alpha \Rightarrow y = \frac{(C-b)}{2} \tan \alpha$$

B = width of furrow

C = distance between adjacent passes

Considering n -th passes, the total vol. of material

$$V = [(n-1)C + B] h_{eff} \quad \text{--- (i)}$$

$$\begin{aligned} h_{eff} &= h_t - h_s - y \\ &= h_t - h_s - \left(\frac{C-b}{2}\right) \tan \alpha \quad \text{--- (ii)} \end{aligned}$$

using eqⁿ. (i) & (ii);

$$V = [(n-1)C + B] \left[(h_t - h_s) - \left(\frac{C-b}{2}\right) \tan \alpha \right]$$

$$\begin{aligned} V &= (n-1)(h_t - h_s)C - \frac{(n-1)C(C-b)}{2} \tan \alpha \\ &\quad + (h_t - h_s)B - \frac{B(C-b)}{2} \tan \alpha \end{aligned}$$

$$V = (\eta-1)(h_t - h_s)C - \frac{(\eta-1)C^2}{2} \tan \alpha + \frac{(\eta-1)bc}{2} \tan \alpha$$

$$+ \cancel{(h_t - h_s)B} - \frac{BC}{2} \tan \alpha + \cancel{\frac{Bb}{2} \tan \alpha}$$

$$dV/dC = 0$$

$$\Rightarrow \frac{dV}{dC} = (\eta-1)(h_t - h_s) - (\eta-1)C \tan \alpha + \frac{(\eta-1)b}{2} \tan \alpha - \frac{B}{2} \tan \alpha$$

-

$$(\eta-1)(h_t - h_s) + \frac{(\eta-1)b}{2} \tan \alpha - \frac{B}{2} \tan \alpha = (\eta-1)C_{opt} \tan \alpha$$

$$C_{opt} = (h_t - h_s) + \frac{b}{2} \tan \alpha - \frac{B}{2(\eta-1)} \tan \alpha$$

$$C_{opt} = (h_t - h_s) \cot \alpha + \frac{b}{2} - \frac{B}{2(\eta-1)}$$

$$\text{As } \eta \rightarrow \infty \Rightarrow \frac{B}{2(\eta-1)} \Rightarrow 0$$

$$C_{opt} = (h_t - h_s) \cot \alpha + \frac{b}{2}$$

$$h_{eff} = h_t - h_s - \frac{(C - b)}{2} \tan \alpha$$

C_{opt}

$$h_{eff} = h_t - h_s - \left[\frac{(h_t - h_s) \cot \alpha + b/2 - b}{2} \right] \tan \alpha$$

$$= h_t - h_s - \left[\frac{(h_t - h_s) - \frac{b}{2} \tan \alpha}{2} \right]$$

$$= \frac{h_t - h_s}{2} + \frac{1}{2} \times \frac{b}{2} \tan \alpha$$

$$h_{eff} = \frac{h_t - h_s}{2} + \frac{b}{4} \tan \alpha$$